

Query Selection for Efficient Black-Box Model Classification

1 Motivation

Classifying black-box generative models requires querying each model with m queries and comparing responses via pairwise energy distances. The total inference cost scales as $m \times n$, so reducing m while maintaining accuracy directly reduces per-model audit cost.

The discriminative factorization estimates per-query signal magnitudes $|\hat{\alpha}_{q,\ell}|$ from a labeled pilot set. We describe three strategies for using this information to select a targeted query set Q^* of size m .

2 Methods

1. Uniform Sampling (baseline). Sample m queries uniformly at random from the pool of M queries. No pilot information is used.

2. Estimated Signal Sampling. Fit a two-component GMM to the SVD loadings $|U_{q,\ell}|$ of the energy tensor E . Queries assigned to the higher-mean component form the estimated signal set $\hat{S}$. Sample m queries uniformly from $\hat{S}$. This exploits zero-set structure but ignores query redundancy.

3. Forward Stepwise Selection. Define a response $z_{ij} = \mathbf{1}[y_i \neq y_j]$ for each model pair and a predictor $E_{q,ij} = \|g(f_i(q)) - g(f_j(q))\|^2$ for each query q . At each step, add the query whose $E_{q,:}$ has the largest partial correlation with the residual of z after projecting out previously selected queries. This is equivalent to Gram-Schmidt orthogonalization on E with respect to z , ensuring each query contributes maximally non-redundant signal.

Stepwise operates on $E \in \mathbb{R}^{M \times \binom{n}{2}}$, already computed during estimation. Cost: $O(m \cdot M \cdot \binom{n}{2})$ — negligible.

3 Experimental Setup

Task	n	M	Description
Motivating	100	200	Sensitive training data
Sys. Prompt	100	200	Covert persuasion bias
RAG	120	200	Unauthorized doc. stores

We use $n_{\text{train}} = 80$ models for estimation and classification, holding out the rest for testing. Query selection uses only the training set. Classification: MDS (8 components) + LDA. Results averaged over 100 random splits.

4 Results

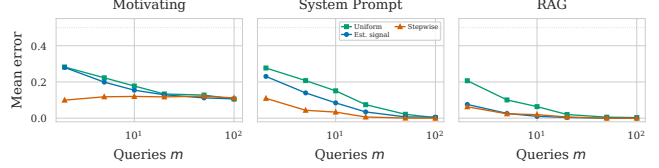


Figure 1: Mean classification error vs. query budget m for three selection strategies across three tasks ($n = 80$). Stepwise selection achieves the same accuracy as uniform sampling with 5–10 $\times$ fewer queries.

Key findings:

- **Stepwise dominates at small m .** At $m=2$: stepwise error is 10% (motivating) and 11% (system prompt), vs. 28% for uniform.
- **Estimated signal is intermediate.** Sampling from $\hat{S}$ consistently outperforms uniform, confirming the GMM partition captures real structure.
- **Practical cost reduction.** On the system prompt task, stepwise at $m=5$ matches uniform at $m=50$ — a 10 $\times$ reduction in per-model query cost.

5 Conclusion

Forward stepwise selection on the energy tensor provides a principled, cheap method for choosing a small query set. Combined with LDA, it matches uniform sampling accuracy with an order of magnitude fewer queries, directly reducing the cost of black-box model auditing.